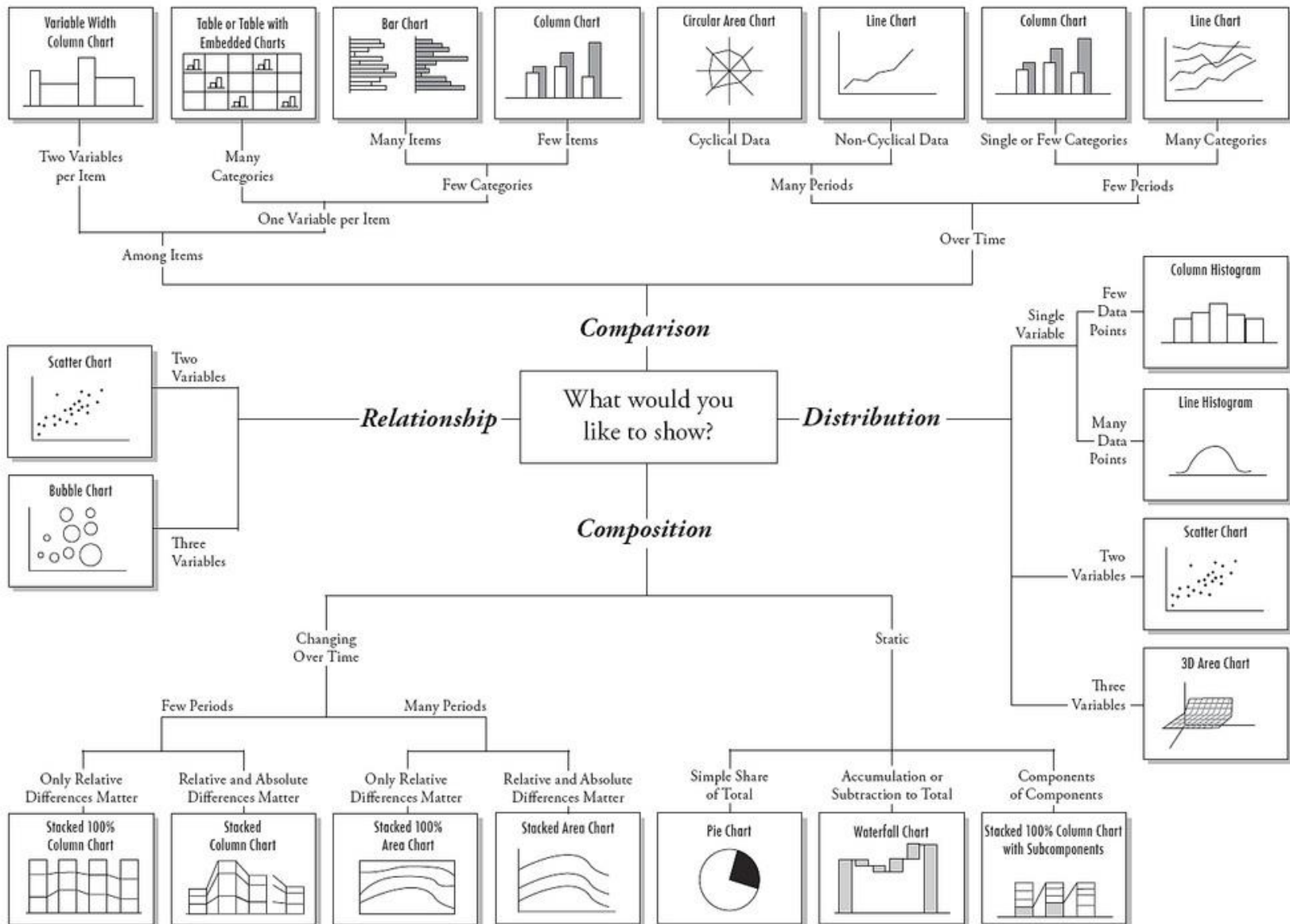


R Plot

How to Represent Data

Chart Suggestions—A Thought-Starter



Plot figure (I)

```
> X11()
> aa<- read.delim("test_data.txt")
> plot(aa$area, aa$price) ## plot area versus price

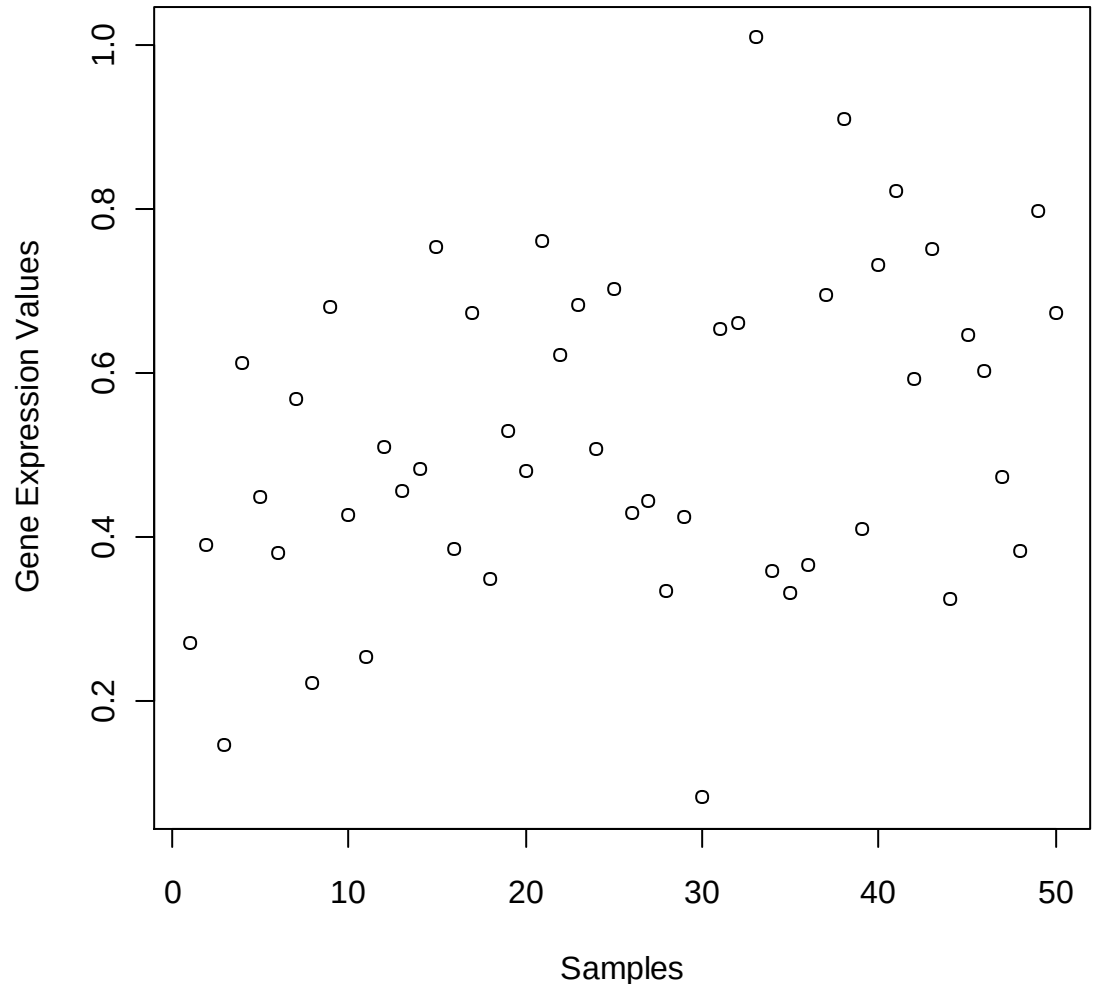
## prepare a graphic device for drawing
## draw on the graphic device
> jpeg(paste(directory, "houseInfo.jpg", sep=""), width = 600,
      height = 480)
> plot(aa$area, aa$price, xlab="area", ylab="price")
> title("house information") ## add title to the plot
> dev.off() ## close the graphic device and complete drawing

> bmp(paste(directory, "houseInfo.bmp", sep=""), width = 480,
      height = 600)
> plot(aa$area, aa$price, xlab="area", ylab="price")
> title("house information")
> dev.off()
```

Plot figure (II)

Scatter Plots of 50 Synthetic Samples

```
plot(data,  
xlab="Samples",  
ylab="Gene Expression Values",  
main=paste("Scatter Plots of",  
length(data),"Synthetic  
Samples",  
sep=" "))
```



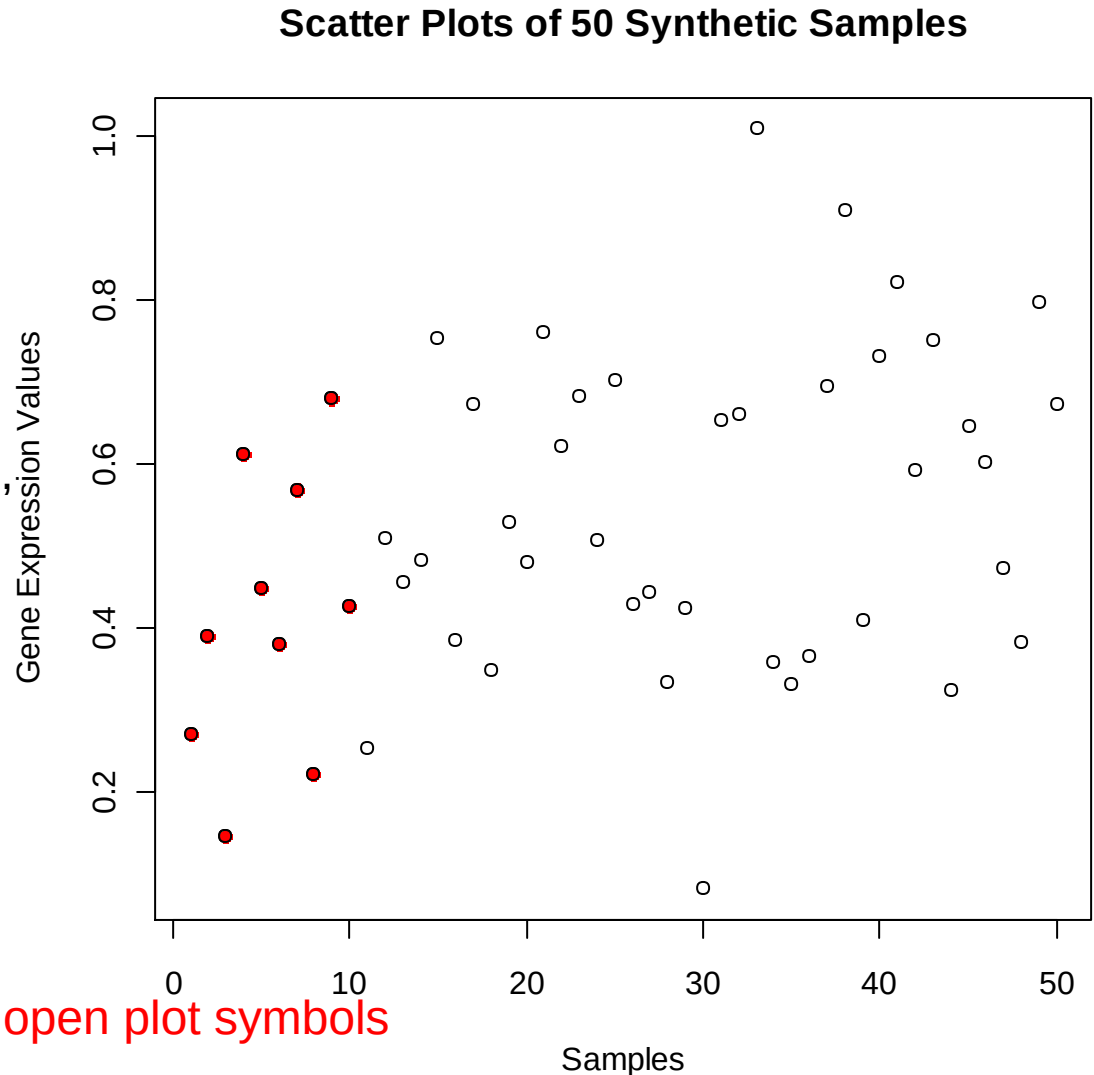
Plot figure (III)

```
plot(data,  
xlab="Samples",  
ylab="Gene Expression Values",  
main=paste("Scatter Plots of",  
length(data),"Synthetic  
Samples",  
sep=" "))
```

```
points(data[11:20], col=1, pch=21,  
bg=2)
```

- How to make data points 11~20 become circle filled in green color?

bg: background (fill) color for the open plot symbols given by pch=21:25.



Graphics symbols

plot symbols : points (... pch = *, cex = 3)

pch="symbol" or pch=k

0		6		12		18		24		0	
1		7		13		19		25		+	
2		8		14		20		*		-	
3		9		15		21		.			
4		10		16		22		o		%	
5		11		17		23		O		#	

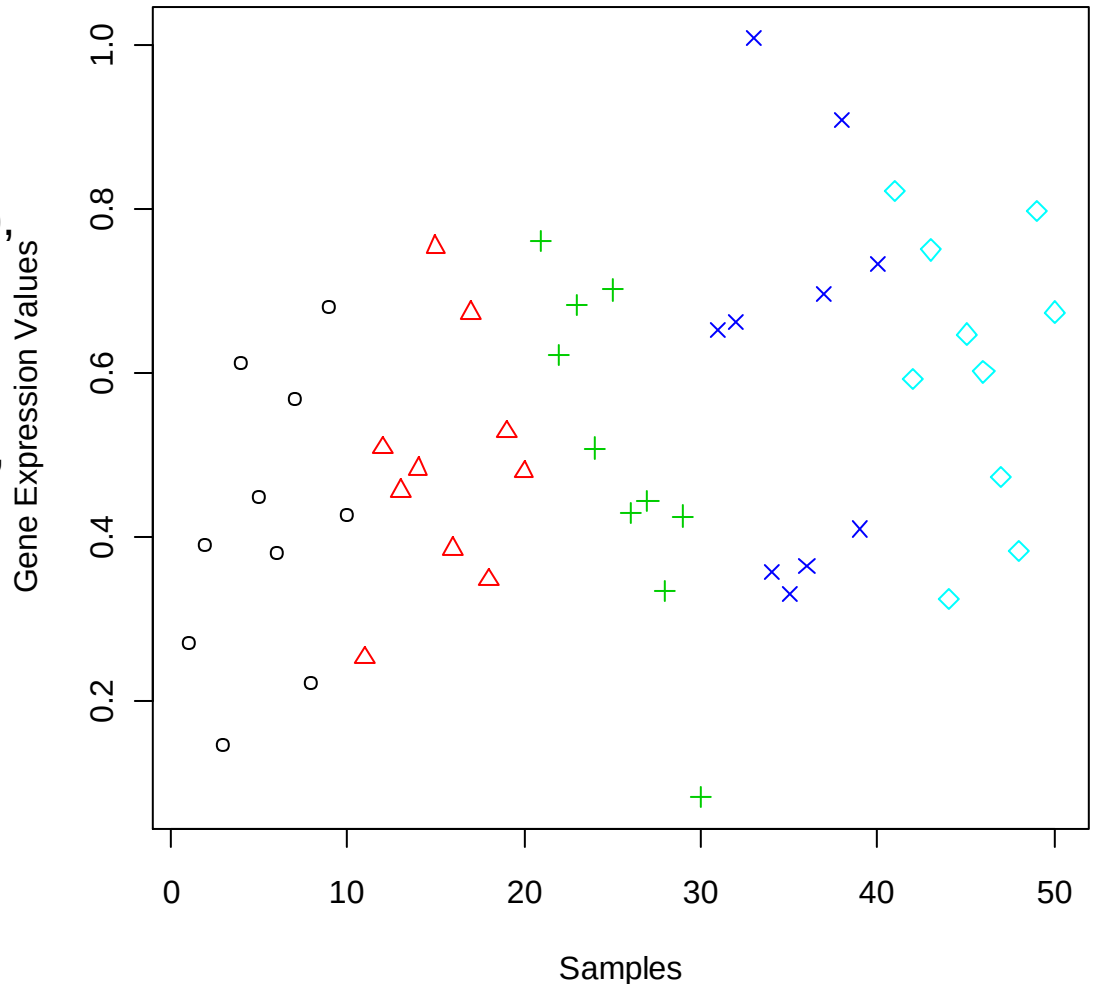
Plot figure (IV)

```
col.lab<-c(rep(1,10),rep(2,10),  
rep(3,10),rep(4,10),rep(5,10))
```

```
plot(data, col=col.lab, pch=col.lab,  
xlab="Samples",  
ylab="Gene Expression Values",  
main=paste("Scatter Plots of",  
length(data),"Synthetic Samples",  
sep=" "))
```

```
## other parameters  
col.axis, col.lab, col.main,  
col.sub
```

Scatter Plots of 50 Synthetic Samples



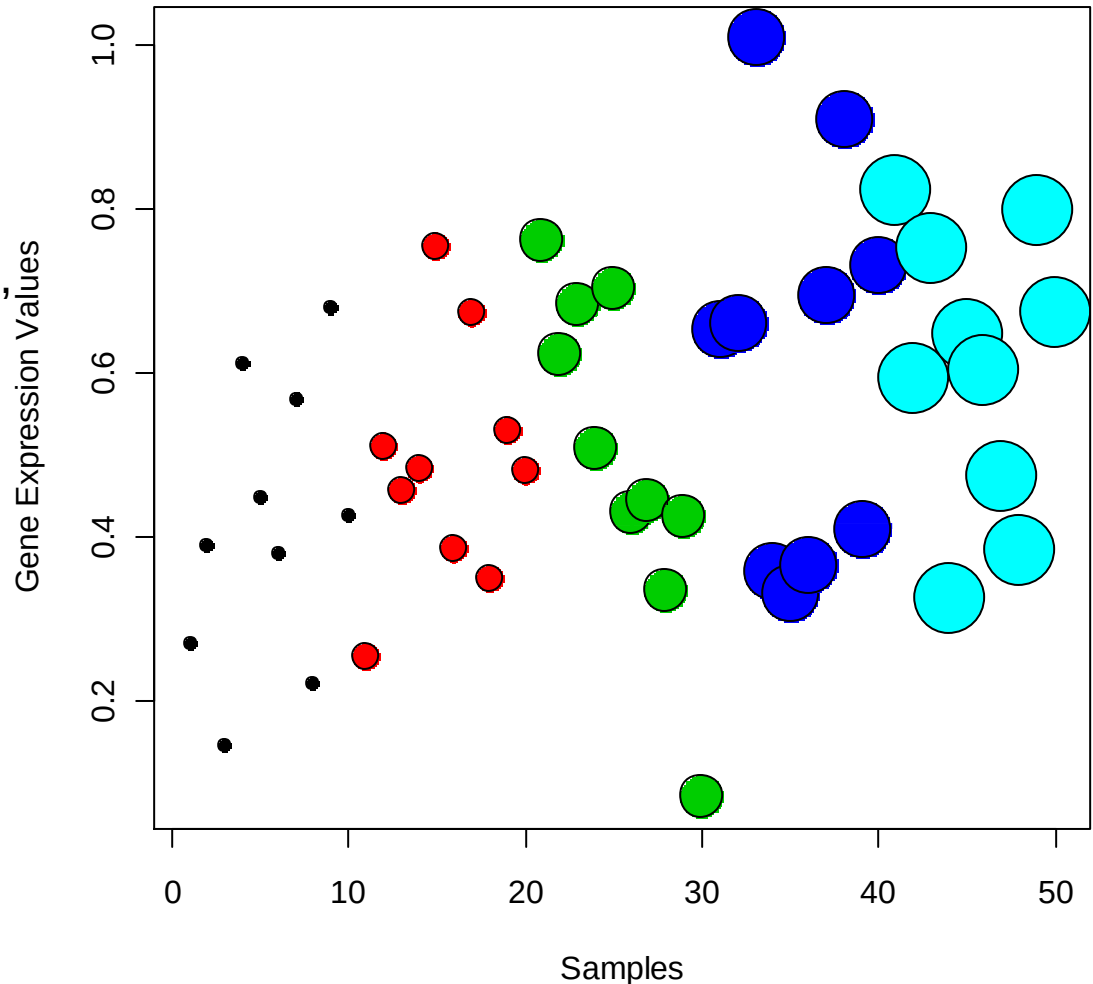
Plot figure (V)

Scatter Plots of 50 Synthetic Samples

```
plot(data, pch=21, bg=col.lab,  
      cex=col.lab, xlab="Samples",  
      ylab="Gene Expression Values",  
      main=paste("Scatter Plots of",  
                  length(data),"Synthetic Samples",  
                  sep=" "))
```

```
## other parameters  
cex.axis, cex.lab, cex.main,  
cex.sub
```

```
font=k, k=1~4  
font.axis, font.lab,  
font.main, font.sub
```



Plot figure (VI)

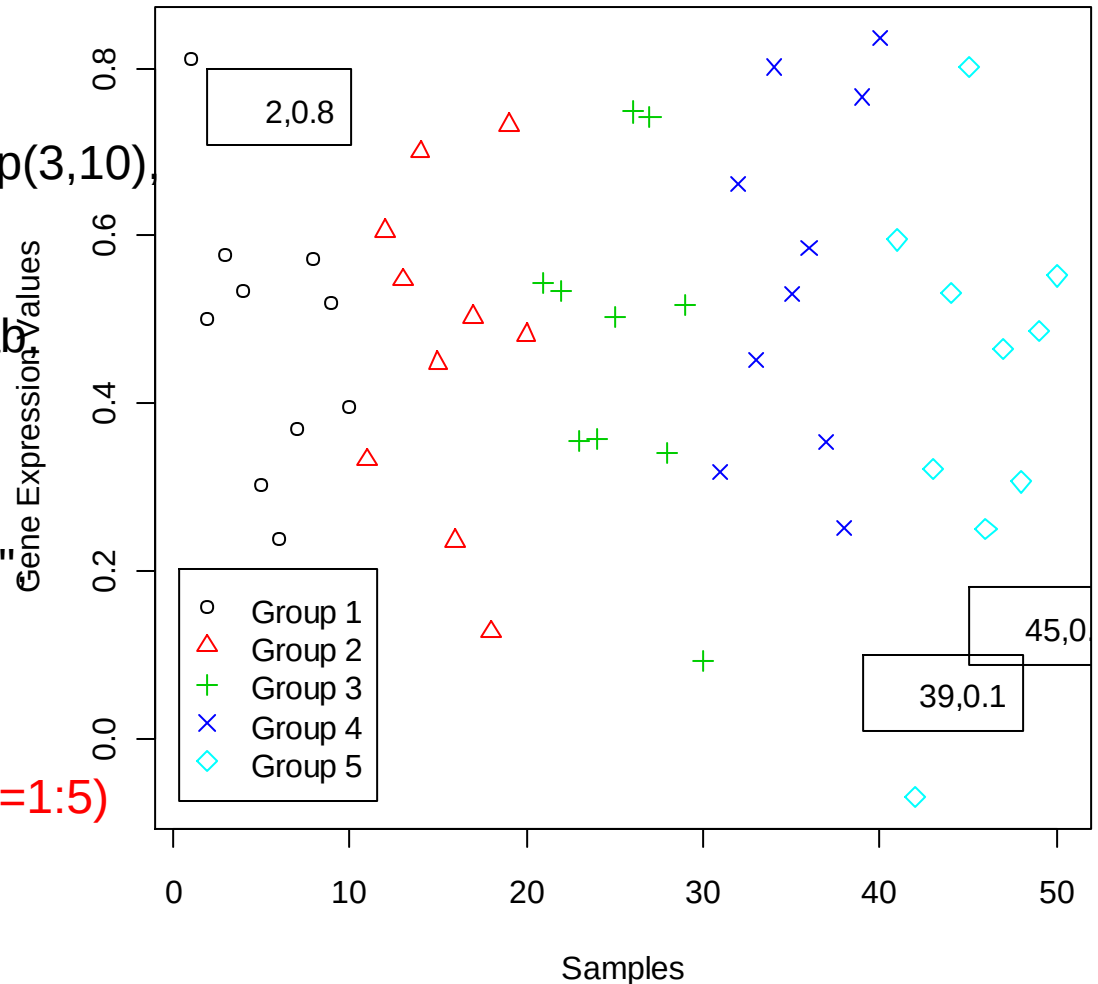
Scatter Plots of 50 Synthetic Samples

```
data<-rnorm(50,0.5,0.2)
```

```
col.lab<-c(rep(1,10),rep(2,10),rep(3,10),  
rep(4,10),rep(5,10))
```

```
plot(data, col=col.lab, pch=col.lab,  
xlab="Samples",ylab="Gene  
Expression Values", main=  
paste("Scatter Plots of",  
length(data),"Synthetic Samples",  
sep=" "))
```

```
legend(locator(1),  
paste("Group",1:5), col=1:5, pch=1:5)  
legend(2,0.8,"2,0.8")  
legend(45,0.18,"45,0.18")  
legend(39,0.1,"39,0.1")
```



Plot figure (VII)

```
x <- 0:64/64
```

```
y <- sin(3*pi*x)
```

```
plot(x, y, type='n')
```

```
legend("bottomright", "(x,y)", pch=1,  
title="bottomright")
```

```
legend("bottom", "(x,y)", pch=1,  
title="bottom")
```

```
legend("bottomleft", "(x,y)", pch=1,  
title="bottomleft")
```

```
legend("left", "(x,y)", pch=1, title="left")
```

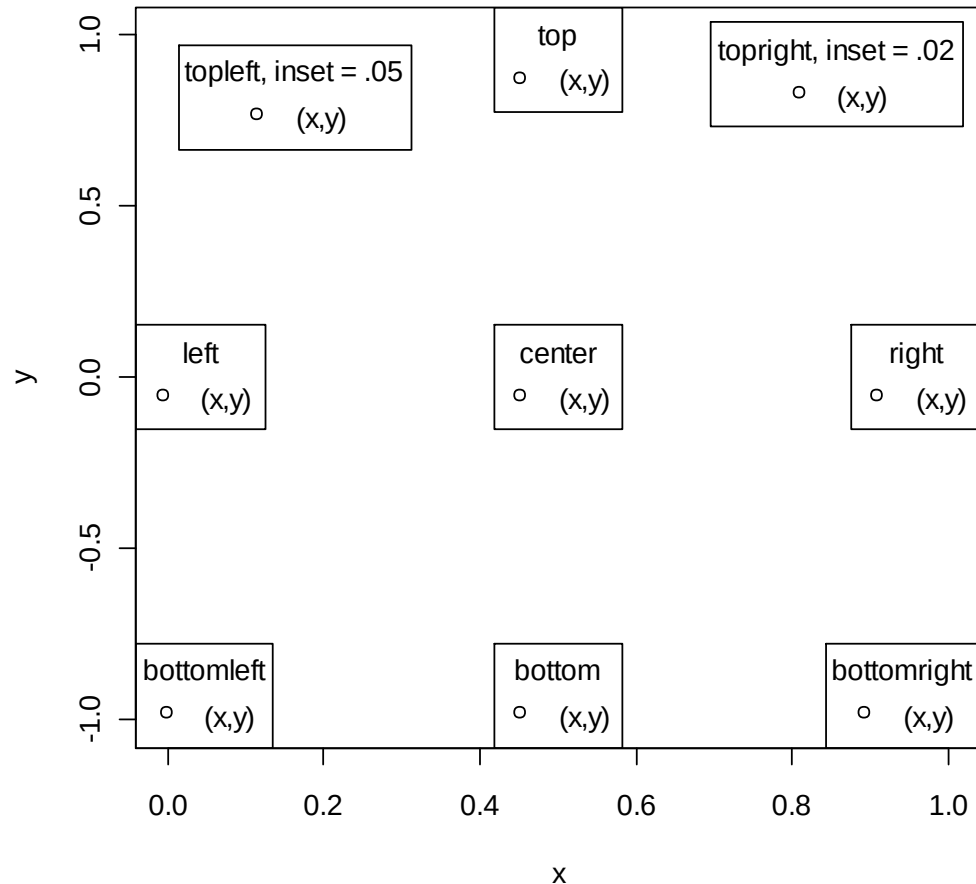
```
legend("topleft", "(x,y)", pch=1,  
title="topleft, inset = .05", inset = .05)
```

```
legend("top", "(x,y)", pch=1, title="top")
```

```
legend("topright", "(x,y)", pch=1,  
title="topright, inset = .02", inset = .02)
```

```
legend("right", "(x,y)", pch=1, title="right")
```

```
legend("center", "(x,y)", pch=1, title="center")
```



Plot figure (VIII)

- `colors()`

```
> length(colors())  
[1] 657  
> colors()[1:10]  
[1] "white"           "aliceblue"       "antiquewhite"    "antiquewhite1"  
[5] "antiquewhite2"   "antiquewhite3"   "antiquewhite4"   "aquamarine"  
[9] "aquamarine1"     "aquamarine2"
```

- `gray.colors()`

```
> gray.colors(5)  
[1] "#4C4C4C" "#888888" "#AEAEAE" "#CCCCCC" "#E6E6E6"
```

- `rainbow()`

```
> rainbow(5)  
[1] "#FF0000FF" "#CCFF00FF" "#00FF66FF" "#0066FFFF" "#CC00FFFF"
```

- `rgb()`

```
> rgb(0, 0, 0, maxColorValue = 255)  
[1] "#000000"  
> rgb(255, 255, 255, maxColorValue = 255)  
[1] "#FFFFFF"
```

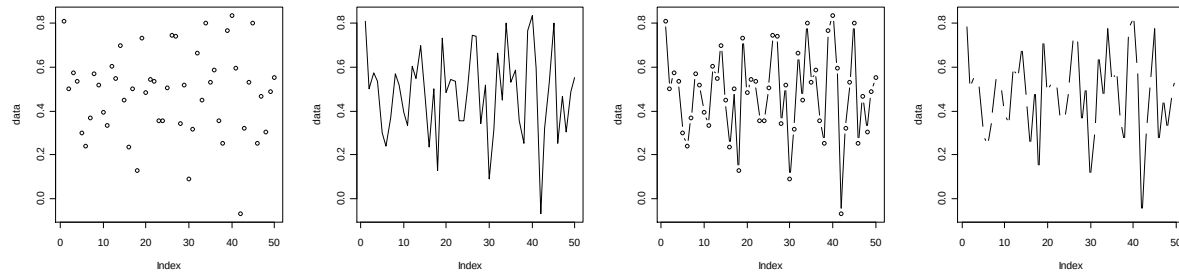
Plot figure (VIII)

- Multiple plots with multiple plot types in a window

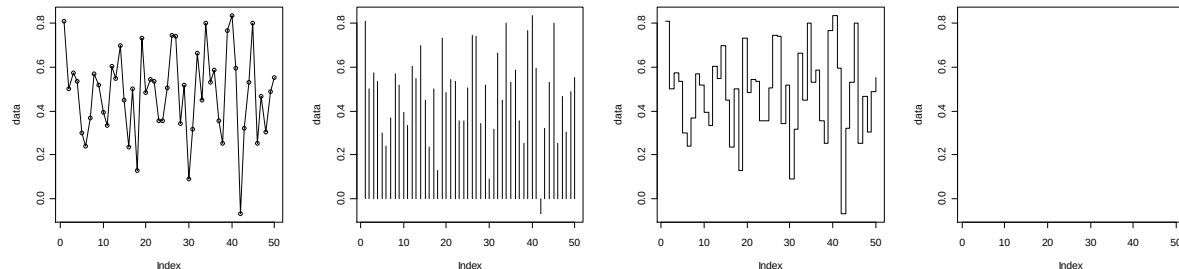
```
data<-rnorm(50,0.5,0.2)
```

```
types<-c("p","l","b","c","o","h","s","n")
```

```
par(mfrow = c(2,4))
for(i in 1:8){
  plot(data, type=types[i])
}
```



```
par(mfcol = c(2,4))
for(i in 1:8){
  plot(data,type=types[i])
}
```



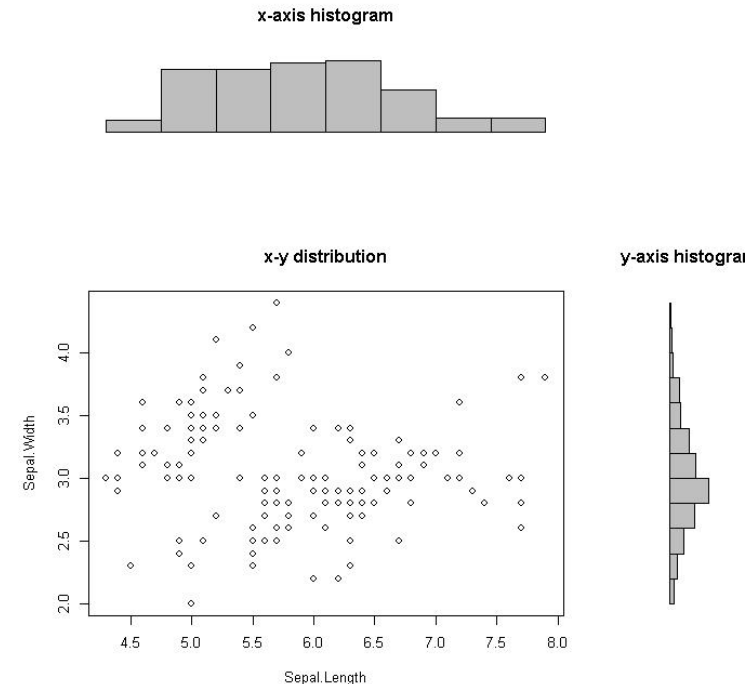
Plot figure (VV)

- Pic with boundary width from bottom, left, up, right (inches)

```
par(mai=c(1, 1.5, 2, 0.5))
```

- Multiple plots in a window by "layout" function

```
> attach(iris)
> x.hist <- hist(Sepal.Length, breaks=10, plot=F)
> y.hist <- hist(Sepal.Width, breaks=10, plot=F)
> top <- max(c(x.hist$counts, y.hist$counts))
> layout(matrix(c(2,0,1,3), 2, 2, byrow=T),
  width=c(4,1), heights=c(1,2))
> plot(Sepal.Length, Sepal.Width, main="x-y
  distribution")
> barplot(x.hist$counts, axes=F, ylim=c(0,top),
  space=0, main="x-axis histogram")
> barplot(y.hist$counts, axes=F, xlim=c(0,top),
  space=0, horiz=T, main="y-axis histogram")
```



Plot figure (VVI)

- Multiple plots in individual windows

```
data<-rnorm(50,0.5,0.2)
```

```
types<-c("p","l","b","c","o","h","s","n")
```

```
for(i in 1:8){  
  X11()  
  plot(data,type=types[i])  
}
```

```
for(i in 1:8){  
  windows()  
  plot(data,type=types[i])  
}
```

- Multiple plots in a PDF file

```
pdf("Multiple_Plots.pdf")
```

```
for(i in 1:8){  
  plot(data,type=types[i])  
}  
dev.off()
```

HW3

- Read the data from the file “SRBCT_train.txt”
 - Normalize data by each row (let values b/w 0~1)
 - Perform scatter plots for Image ID “770394”, “814260”, and “491565”
 - Perform scatter plots for Image ID “770394” vs “236282”, and “812105” vs “784224”
 - Diff. colors, symbols & legends for samples in diff. categories

Hint:

```
data_SRBCT<-read.delim("SRBCT_train.txt")
data1<-as.matrix(data_SRBCT[,-c(1:2)])
name1<-data_SRBCT[,1]
data1_nor<-t(apply(data1,1,nor))
plot(data1_nor[which(name1=="770394"),])
```

```
nor<-function(x){
  min.x<-min(x)
  max.x<-max(x)
  (x-min.x)/(max.x-min.x)
}
```